TECHNICAL PRESENTATION

Team no. 3

12. Big Data Analytics in Cloud Computing

CSA1519 – Cloud Computing and Big Data Analytics

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Abstract

- Big data → massive, diverse and rapidly generated datasets
- Cloud platforms → scalable storage and on-demand processing
- Distributed systems → process data in parallel across clusters
- Analytics → descriptive, predictive and real-time insights
- Focus → architecture, platforms, business value and applications

Aim and Objective

- Aim → explore big data analytics using cloud computing
- Understand cloud-based data processing architectures
- Examine storage, distributed processing and analytics platforms
- Identify business value, scalability and cost benefits
- Explore security, governance and data management challenges
- Study real-world applications across industries

Introduction

- Big data is characterized by volume, velocity and variety
- Cloud computing provides resources over the internet on demand
- Cloud storage supports large, growing datasets
- Distributed processing handles data in parallel
- Analytics converts raw data into actionable insights

Core Technical Explanation

- Data sources → apps, sensors, transactions and social media
- Ingestion → batch or streaming data enters the cloud
- Storage → data lakes, object stores and cloud warehouses
- Processing → distributed engines clean and transform data
- Analytics → dashboards, machine learning and prediction
- Key concerns → quality, privacy, security and governance

Existing System

- Traditional systems often rely on fixed on-premises capacity
- Large datasets may require costly hardware upgrades
- Scaling storage and compute can take time
- Separate tools can create data silos and complex workflows
- Cloud platforms offer elastic resources and managed services
- Challenges → cost control, security and vendor dependence

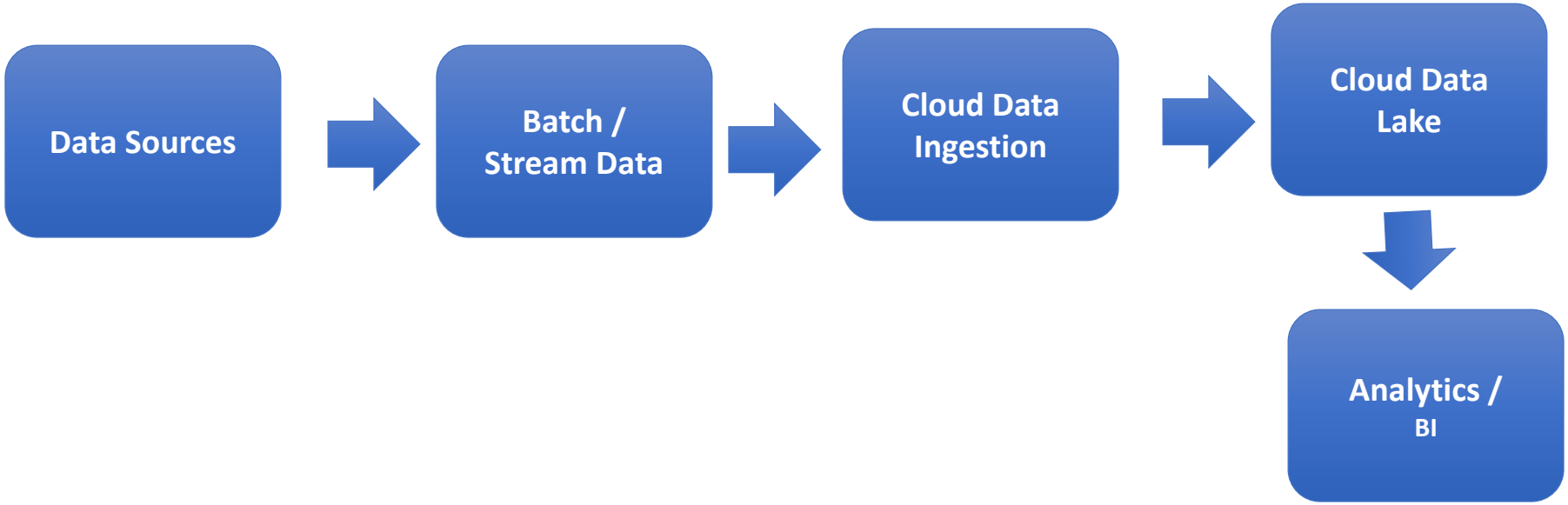
Proposed System

- Collect structured, semi-structured and unstructured data
- Ingest data through batch pipelines or streaming services
- Store raw data in a scalable cloud data lake
- Process data using distributed compute frameworks
- Apply SQL analytics, machine learning and visualization
- Monitor access, quality, performance and cloud spending

Overall System Architecture Design

- Sources → applications, business systems, devices and logs
- Ingestion layer → batch loads and real-time streams
- Storage layer → cloud object store / data lake / warehouse
- Processing layer → distributed compute and data preparation
- Analytics layer → BI dashboards, ML models and reports
- Security layer → IAM, encryption, governance and monitoring

Proposed Methodology and Algorithm



1. Identify business question and required data
2. Ingest data using batch or streaming pipelines
3. Validate, clean and transform incoming datasets
4. Store data in a scalable and governed cloud repository
5. Process data in parallel and run analytics models

Real-world use case in cloud computing

Retail → demand forecasting, recommendations and customer trends

Healthcare → population insights and clinical data analysis

Banking → fraud detection and risk analytics

Manufacturing → predictive maintenance and quality monitoring

Transport → route optimization and fleet analytics

Smart cities → traffic, energy and environmental monitoring

Conclusion

Cloud computing makes large-scale analytics more accessible

Distributed processing supports fast, scalable data workloads

Cloud platforms integrate storage, processing and machine learning

Business value includes better decisions and operational efficiency

Security, governance and cost management remain essential

Successful projects align architecture with business needs

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